



Renaud Mariana

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Summary

With over 15 years of experience in programming, operating and testing SaaS software and storage systems; I'm keen to exploit my knowledge further, helping with local AI inference (RAG) driven applications

WORK EXPERIENCE

Santevet | Online bookings

France - remote 

23 - till present

Key Responsibilities

- Currently working on local LLMs
 - Build tools on top of local LLMs with llama.cpp.
- Enforce software robustness of a Phoenix online bookings SaaS:
 - Fix several critical bugs.
 - Fix aws instance sizes and wrong linux settings.
 - Solving some front bugs.
- Bring observability:
 - Add high resolution metrics (user metrics).
 - Add monitoring with Prometheus / PromQL.
 - Build salient Grafana Dashboards.
- Stripe: add a card payment method to Santevet products and migrate our Stripe api.
- Implement a garbage collector for CQRS SQL data.

Github

- Commanded (CQRS/ES): fix an OOM bug present for years in the commanded dependency;
- Prometheus high resolution metrics : create a small memory footprint prometheus counter able to track different activities for all users, used intensively by our Analytics
- local llm scripts

Environment

- AWS - Postgres - Phoenix - CQRS/ES - EventStore - Commanded - Elasticsearch - Stripe API - PromQL - Broadway - Rust

Motorola Solutions

Glostrup, Denmark - on-site 

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Key Responsibilities

- Devops for a cloud-based telecom product that connects Land Mobile Radio systems over the internet;
- **Reliable software programming - kubernetes devops**
- Software: write new kubernetes microservices (service-level requirement, ..), associated tests and deployed the services in all regions;
The role included tasks like writing / running the deployment pipelines, the CI/CD integration, templates, helm chart, monitoring, logging and observability
- SRE tasks: successfully implement a **SLA measurement** microservice, join on-call rotation, run post-mortem, improve reliability

Environment

- Kubernetes - Helm - Docker - Elixir - Elk - Azure Pipelines - Telemetry - PagerDuty - Prometheus - Continuous Integration - Rabbitmq - Grafana - Redis - Phoenix - GraphQL.

Kuantic | Lead

Sofia-Antipolis, France

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Key Responsibilities

- Fleet Management SaaS – Technical interface for Stellantis
- Architectural tasks:
 - Modernizing the software development and deployment process
 - Backend code refactoring
 - **Reshape the fleet management API into a more general data broker API** (concepts + software)
- Reliability: improve the stack reliability and extensibility with the help of a MQ
- Big-data: Transform a fleet traffic model into a data broker business model: mapReduce pipelines, configurable customers queues, third-party export pipelines
- Tooling: add tools like Elastic Search and Elixir Dashboards
- Troubleshoot some issues
- Performance: improve overall performance with some caching
- Side project: develop and maintain a SaaS based on Phoenix / LiveView

Github

- A [local key-value cache](#);
- A [nif cache](#) for openstreetmap.

Environment

- NoSQL · MapReduce · Erlang · Distributed Systems · RabbitMQ · Broadway · Elixir · SQL · Docker · Kubernetes · Phoenix · Elastic Search

Scaleway | R&D

Paris, France

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Key Responsibilities

- Distributed Cloud Storage – R&D – Devops
 - From scaleway requirements, define a distributed and fault-tolerant storage architecture
 - Implement a flexible, lightweight protocol layer ([nbd](#)) in erlang (where it shines), allowing for example to relocate customers storage while in use
 - *Linux, Live release upgrade, app profiling, riak LRU cache, block dependency graph, users volume live relocation, riak node crash recovery, data flow performance analysis, commodity computing, webmachine rest api*
- **AI fraud detection system**
 - Ownership of the fraud (rent servers abuses) decision engine design
 - Build the features and python workflow (SQLAlchemy, Rest API)
 - The built AI fraud detection is used in production and is fully automated for the support team, probably the first AI agent ever used in the company
 - *Scikit-learn, Python, SQLAlchemy, Postgresql*
- Online SAS san servers as [IOT](#) (POC) for live snapshots and statistics data collection
topics: qemu-nbd, iot
- [Erlang distributed testing](#) of AWS s3
topics: basho_bench, tsung

Github

- a [spatio-temporal](#) logger.

Environment

- Scikit-Learn · PostgreSQL · NoSQL · Storage Area Network (SAN) · Machine Learning · Erlang · Python · SQLAlchemy · Distributed Systems · Quickcheck · Docker · Qemu · Linux Kernel

Meetic, +500 people

Paris, France

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Key Responsibilities

- Distributed chat system – main backend developer
- Maintainer of the messaging stack (Erlang/ [Ejabberd](#) **+200K simultaneous users**)
- Architect and code new requirements for the Meetic chat system
- Add the xmpp protocol bosh extension to the backend
- Package, benchmark and R&D work in erlang:
 - Development of a riak-jabberd roster (contacts)
 - Development of a game framework as a cluster extension for Ejabberd to support simultaneously ten of thousands of human interactions stored in Riak
 - A realtime 3D supertracker to monitor fraudulent chat messages
 - Benchmark RabbitMQ for messages logging
- Side project: rewrite in Erlang a C++ biological follicular/ovarian growth simulator I built for INRA, (ATT/C++ coroutines), my first project in Erlang ❤️
- Side projects: open-source a real-time extension for erlang, allowing to schedule hard real-time messages, illustrated with two [N2O](#) projects:
 - An Erlang multi-player midi sequencer
 - An interactive physics simulator

Github

- A [real-time NIF](#) extension for Erlang, allowing to schedule real-time messages.

Consultant

embedded software

Paris, France

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• Embedded programming – Philips / LG / Sagem mobiles

- Based on the 18Crypt profile, architect of the NXP mobileTV CAS feature (conditional access); this multi-site project involved people from the Netherlands, India and Ireland (S3)
- Feature leader of Puma phone media manager, specify & code the real-time requirements (apollo, js)
- Development of a new video player automaton in C for Sagem Mobile
- Build a host-based verification system tool to ensure the middleware correctness, this tool uses logic programming techniques on a host to drive the tests on the embedded system
- Android devices: build a verification tool to test low-level Android java hooks (thread, java fields / methods, state machines)

• Embedded programming – IPTV [STB](#)

- Add multi-threading ([EMM/ECM](#) and Tuner) for the Philips STB DLI (Device Layer Interface); develop a porting layer for PVR services based on ST40 crypto security
- As a field engineer at BYTEL for the BBOX IPTV Product, investigates Technicolor stb Middleware
- Rewrite in Erlang my massively parallel C++ coroutines simulator of biological processes (based on Cox Renewal Theory statistics model)

• DVB-H Mobile TV – Orange

- Head of the developments for the next generation SIM & KDA conditional access

	- Architecture of a poc (2 patents) server - mobile - card , mp4 encryption schemes, licence packaging - Manage the relationships with major smart card suppliers, video software vendors and third-party software companies - Semi-formal testing of the sim software using logic programming techniques , customize VLC to decode the rtp h264 ismacryp stream
CP8 Transac <i>Louveciennes, France</i> July 97 - June 2002	Project leader of CP8 first javacard (3 patents including 2 US) - Did the coordination between INRIA, the mask and the CP8 R&D teams. This project included a virtual machine simulator along with other tools like secure loader or bytecode optimization programs and benchmarks suite software - R&D: wrote many javabytecode optimizations and javacard demos like a mastermind solver for benchmark - Replace the old proprietary VB tools with web applications for the new HTTP smartcard interface (one patent) - Technical consultancy for a French bank (Caisse d'Epargne, B0' application) Secure protocols for e-commerce, Carte Bleue  - Port the French banking protocol (B0') for javacard - Add multi-threading programming for the SimToolkit (3 patents have been filled) - Proof of concept of a sms-based E-purse transaction system (Proton smartcard and server) - Open-source the Prolog <u>PC/SC interface</u> , used to run semi-formal tests in logic programming

EDUCATION

Groupe Ecoles Centrales <i>Centrale Méditerranée, France</i>	- Graduated in Computer Engineering Minor: Industrial IT and embedded systems - Competitive classes preparing students to the best engineering schools ("Les grandes Ecoles")
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Computer SKILLS

RAG development:	Elixir, Nx, Bumblelebee, ollama, langchain and Jina
Systems:	SaaS, Kubernetes / Docker, local LLM, Azure pipelines CI/CD, low latency Erlang/OTP, Elixir ecosystem, Phoenix, Tailwind-css, github actions, riak (nosql), linux kernel (bloc storage), graphQL, sqlite, Postgresql, SQLAlchemy, Tsung, redis, C++ Coroutines
Prediction - Observability:	RAG - local LLM programming - PromQL - High Resolution Counters
Programming languages:	Erlang / Elixir, C/C++, Python, JavaScript prototype, java-bytecode, SQL, ...
Methodologies:	DevOps best practices (Motorola)
Protocols:	Nbd, aws, xmpp, iptv, TR-069, mpeg-4 streaming (RFC 3984), ISMACryp 1.0, Etsi-Tetra
Embedded:	Linux, Nucleus, RTKI, VME, JavaCard and Android
Smart Cards:	EMV, SIM Card, B0' Banking Application, E-Purse, OpenPlatform

Personal PROJECTS

Prometheus:	Create a small memory footprint <u>telemetry_metrics_prometheus_core</u> counter able to track millions of different counters
LiveView-based SaaS:	Develop and maintain a SaaS based on Phoenix / LiveView / live SVG
Superls:	A multi-volumes, files tokenizer, indexer and <u>search engine</u>
Caching systems:	Small libraries on top of a NOSQL database
Spatio-temporal logger:	A tiny library to debug concurrency in distributed systems, both in time and space
Hardware clock:	ALSA-hardware-based clock in C (NIF) for real-time message processing in Erlang
Ejabberd Game framework:	Framework for multi-player games on top of Riak and Ejabberd